

ORDER AND ORDER ITEMS MANAGEMENT

Project Title: IKEA – E-Commerce Database Management System

1.Database design

1.1 Orders Table

Attribute	Data Type	Key	Description
Order_ID	INT	PK	Unique order ID
Customer_ID	INT	FK	References Customer
Order_Date	DATE	-	Order date
Total_Amount	DECIMAL(10,2)	-	Total amount
Order_Status	VARCHAR(30)	-	Placed/Shipped/Delivered
Shipping_Address	TEXT	-	Shipping address
Delivery_Date	DATE	-	Delivery date

1.2 Orders_items Table

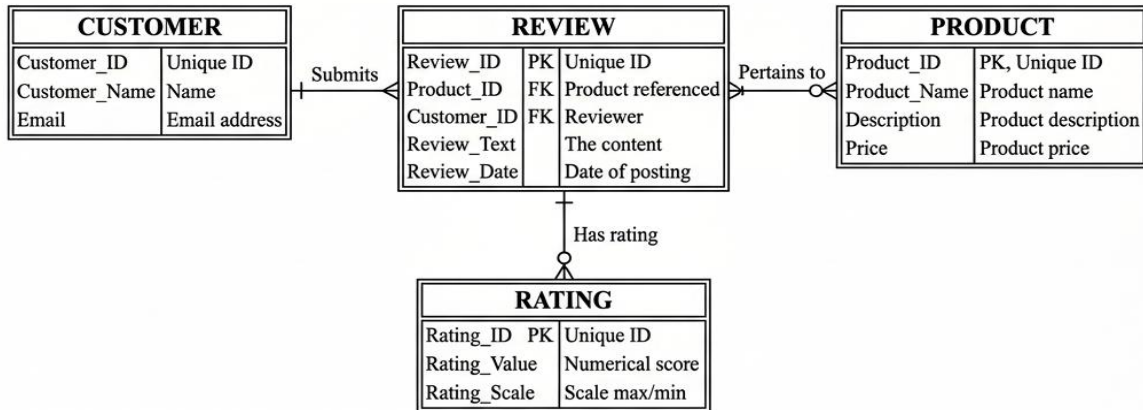
Attribute	Data Type	Key	Description
Order_Item_ID	INT	PK	Unique order item ID
Order_ID	INT	FK	References Orders
Product_ID	INT	FK	References Product
Quantity	INT	-	Quantity
Unit_Price	DECIMAL(10,2)	-	Unit price
Discount	DECIMAL(5,2)	-	Discount
Total_Price	DECIMAL(10,2)	-	Total price

2. Table Relationships

FIGURE 1: E-Commerce Database Relationships (Customer and Product)

2. TABLE RELATIONSHIPS:

2.1 SCHEMA DESIGN:



2.2 RELATIONSHIP SUMMARY:

CUSTOMER - REVIEW RELATIONSHIP (1:M)	
Attributes	Description
Customer_ID	Found in both tables. Links customer to review. Each review belongs to exactly one customer, but one customer can write many reviews.
PRODUCT - REVIEW RELATIONSHIP (1:M)	
Attributes	Description
Product_ID	Found in both tables. Links review to product. Each review is about one product, but one product can have many reviews.
REVIEW - RATING RELATIONSHIP (1:M)	
Attributes	Description
Rating_ID	Found in both tables. A review can generate a detailed numerical rating structure, or be linked to a rating matrix. One review can contain multiple individual ratings (e.g., quality, usability, shipping speed).

3. Table Creation

3.1 Create Orders Table

```
SQL> CREATE TABLE Orders (  
2   Order_ID INT PRIMARY KEY,  
3   Customer_ID INT,  
4   Order_Date DATE,  
5   Total_Amount DECIMAL(10,2),  
6   Order_Status VARCHAR(30),  
7   Shipping_Address VARCHAR(255),  
8   Delivery_Date DATE  
9 );
```

Table created.

3.2 Create Order_Items Table

```
SQL> CREATE TABLE Order_Items (  
  2     Order_Item_ID INT PRIMARY KEY,  
  3     Order_ID INT,  
  4     Product_ID INT,  
  5     Quantity INT,  
  6     Unit_Price DECIMAL(10,2),  
  7     Discount DECIMAL(5,2),  
  8     Total_Price DECIMAL(10,2)  
  9 );  
  
Table created.
```

3.3 Order Data Insertion

```
SQL> INSERT INTO Orders VALUES  
  2 (1001, 1, DATE '2026-09-01', 12999.00, 'Delivered', 'Chennai, Tamil Nadu', DATE '2026-09-05');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Orders VALUES  
  2 (1002, 2, DATE '2026-09-03', 6999.00, 'Shipped', 'Bengaluru, Karnataka', DATE '2026-09-08');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Orders VALUES  
  2 (1003, 3, DATE '2026-09-05', 5999.00, 'Placed', 'Hyderabad, Telangana', NULL);  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Orders VALUES  
  2 (1004, 4, DATE '2026-09-07', 4999.00, 'Delivered', 'Mumbai, Maharashtra', DATE '2026-09-11');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Orders VALUES  
  2 (1005, 5, DATE '2026-09-10', 3999.00, 'Shipped', 'Pune, Maharashtra', DATE '2026-09-14');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Orders VALUES  
  2 (1006, 6, DATE '2026-09-12', 1499.00, 'Placed', 'New Delhi', NULL);  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Orders VALUES  
  2 (1007, 7, DATE '2026-09-15', 1999.00, 'Delivered', 'Noida, Uttar Pradesh', DATE '2026-09-19');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Orders VALUES  
  2 (1008, 8, DATE '2026-09-18', 8999.00, 'Shipped', 'Kolkata, West Bengal', DATE '2026-09-23');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Orders VALUES  
  2 (1009, 9, DATE '2026-09-20', 2499.00, 'Placed', 'Ahmedabad, Gujarat', NULL);  
  
1 row created.
```

4. Order items Data Insertion

```
INSERT INTO Order_Items VALUES
2 (1, 1001, 102, 1, 12999.00, 0.00, 12999.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Items VALUES
2 (2, 1002, 104, 1, 6999.00, 0.00, 6999.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Items VALUES
2 (3, 1003, 105, 1, 5999.00, 0.00, 5999.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Items VALUES
2 (4, 1004, 101, 1, 4999.00, 0.00, 4999.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Items VALUES
2 (5, 1005, 106, 1, 3999.00, 0.00, 3999.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Items VALUES
2 (6, 1006, 107, 1, 1499.00, 0.00, 1499.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Items VALUES
2 (7, 1007, 108, 1, 1999.00, 0.00, 1999.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Items VALUES
2 (8, 1008, 109, 1, 8999.00, 0.00, 8999.00);

1 row created.

SQL>
SQL> INSERT INTO Order_Items VALUES
2 (9, 1009, 103, 1, 2499.00, 0.00, 2499.00);

1 row created.
```

5. Order Modification Operations

5.1 Modify Order Amount

```
SQL> UPDATE Orders
2 SET Total_Amount = 13499.00
3 WHERE Order_ID = 1001;

1 row updated.

SQL>
SQL> COMMIT;

Commit complete.
```

6.1 Modify Order Date:

```
SQL> UPDATE Orders
  2  SET Order_Date = DATE '2026-09-02'
  3  WHERE Order_ID = 1001;

1 row updated.

SQL>
SQL> COMMIT;

Commit complete.
```

7. Order_Items Modification

7.1 Modify Order Item Quantity

```
SQL> UPDATE Order_Items
  2  SET Quantity = 2,
  3      Total_Price = 25998.00
  4  WHERE Order_Item_ID = 1;

1 row updated.

SQL>
SQL> COMMIT;

Commit complete.
```

7.2 Modify Order Item Price

```
SQL> UPDATE Order_Items
  2  SET Unit_Price = 13500.00
  3  WHERE Order_Item_ID = 1;

1 row updated.

SQL>
SQL> COMMIT;

Commit complete.
```

8. Customer Order History Report

```
SQL> SELECT
  2   o.Order_ID,
  3   o.Customer_ID,
  4   o.Order_Date,
  5   o.Total_Amount,
  6   o.Order_Status,
  7   oi.Product_ID,
  8   oi.Quantity,
  9   oi.Unit_Price,
10   oi.Total_Price
11 FROM Orders o
12 JOIN Order_Items oi
13 ON o.Order_ID = oi.Order_ID
14 ORDER BY o.Order_Date;

ORDER_ID CUSTOMER_ID ORDER_DATE TOTAL_AMOUNT ORDER_STATUS
-----
PRODUCT_ID QUANTITY UNIT PRICE TOTAL PRICE
-----
1001      1 02-SEP-26      13499 Delivered
102      2      13500      25998
1002      2 03-SEP-26      6999 Shipped
104      1      6999      6999
1003      3 05-SEP-26      5999 Placed
105      1      5999      5999

ORDER_ID CUSTOMER_ID ORDER_DATE TOTAL_AMOUNT ORDER_STATUS
-----
PRODUCT_ID QUANTITY UNIT PRICE TOTAL PRICE
-----
1004      4 07-SEP-26      4999 Delivered
101      1      4999      4999
1005      5 10-SEP-26      3999 Shipped
106      1      3999      3999
1006      6 12-SEP-26      1499 Placed
107      1      1499      1499

ORDER_ID CUSTOMER_ID ORDER_DATE TOTAL_AMOUNT ORDER_STATUS
-----
PRODUCT_ID QUANTITY UNIT PRICE TOTAL PRICE
-----
1007      7 15-SEP-26      1999 Delivered
108      1      1999      1999
1008      8 18-SEP-26      8999 Shipped
109      1      8999      8999
1009      9 20-SEP-26      2499 Placed
103      1      2499      2499

ORDER_ID CUSTOMER_ID ORDER_DATE TOTAL_AMOUNT ORDER_STATUS
-----
PRODUCT_ID QUANTITY UNIT PRICE TOTAL PRICE
-----
1000     10 22-SEP-26      799 Delivered
110      1      799      799

10 rows selected.

SQL> |
```

9 Customer-wise Order Summary

```
SQL> SELECT
  2   Customer_ID,
  3   COUNT(Order_ID) AS Total_Orders,
  4   SUM(Total_Amount) AS Total_Amount
  5 FROM Orders
  6 GROUP BY Customer_ID
  7 ORDER BY Customer_ID;

CUSTOMER_ID TOTAL_ORDERS TOTAL_AMOUNT
-----
1          1          13499
2          1           6999
3          1           5999
4          1           4999
5          1           3999
6          1           1499
7          1           1999
8          1           8999
9          1          2499
10         1           799

10 rows selected.
```